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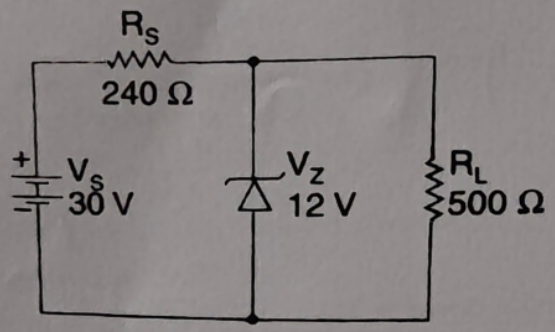
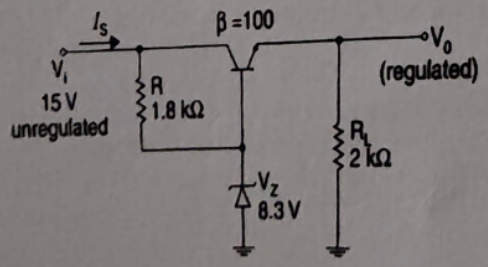
EEA-103
Examination –Dec.- 2023
B.Tech. II Sem : Electrical Engineering
Basic Electronics

Time : 2 Hrs

Max. Marks : 60
Min. Marks : 19

Note: Total number of questions are 05. All Questions are compulsory. Each Question has 4 parts (a, b, c, d). Part a & b are compulsory while Part c & d has internal Choice. Assume missing data, if any.
Word limit be observed as follows:
Part a & b – Objective type/Fill in the blanks/One sentence answer.
Part c – Max 100 words and Part d – Max 400 words.
Word limit NOT to be followed for diagram, numerical, derivation.

			Marks	BL	CO
Q.1	(a)	(i) When a diode is reversed biased, it is equivalent to 1. An OFF switch 2. An ON switch 3. A high resistance 4. None of the above	02	1	C01
		(ii) A LED emits visible light when its 1.P-N junction is reversed-biased 2. Depletion region widens 3. holes and electrons recombine 4. P-N junction becomes hot.		1	CO1
	(b)	A two-diode rectifier will not operate without a _____.	02	1	CO1
	(c)	What is a PIN diode and its applications.	03	1,2	CO1
		OR			
		Draw the V-I characteristics of a tunnel diode and indicate the useful region in the curve	03	2,3	CO1
	(d)	Explain Light emitting diode. Discuss Seven-segment display	05	1,2	CO1
		OR			
		Determine te germanium PN junction diode current for the forward bias voltage of 0.22 V at room temperature 25^0 C with reverse saturation current equal to 1m A . Take η=1.	05	3	CO1
Q.2	(a)	(i) The element that has the biggest size in a transistor is …….. 1. collector 2. base 3. emitter 4. collector-base-junction	02	1	CO2

Q.5	(a)	(i) In a Zener diode shunt voltage regulator, the diode regulates so long as it is kept in..... 1. Forward 2. Reverse 3. Loaded 4. unloaded	02	1	CO5
		(ii) What are the disadvantages of the series regulators? 1. Higher current stability 2. Simple to design 3. Inefficient 4. None of the above		1	
	(b)	_____ are the building blocks of regulated DC power supply	02	1,2	CO5
	(c)	Explain the zener diode shunt regulator?	03	1,2	CO5
		OR			
		Figure shows the circuit of a zener diode shunt regulator Find the load voltage, voltage drop across series resistance.  	03	3	CO5
	(d)	List the types of Voltage Regulators? Explain transistor series Regulator?	05	1,2	CO5
		OR			
		Calculate the output voltage and Zener current in the regulator circuit shown in fig.  	05	3	CO5

Bloom's Taxonomy Levels (BL): 1-Remembering, 2- Understanding, 3- Applying, 4- Analysing, 5- Evaluating, 6-Creating

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